

Software Requirement Specification (SRS)

User Requirements

- Authentication and Session Management
 - Resume Upload and Processing
 - AI Resume Analysis and Feedback
 - User Dashboard and Analytics
 - Admin Panel Management
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System Requirements

1. Authentication and Session Management

1.1 New users can create an account by entering their name, email address, password, and confirm password.

1.2 The system will check whether the password and confirm password match. If they do not match, an error message saying "**Passwords do not match**" will be displayed without reloading the page.

1.3 Registered users and administrators can log in using their email address and password.

1.4 When the user clicks the **Login** button, the system will verify the login information from the database and create a secure login session with the user's role (Admin or User).

1.5 If the email or password is incorrect, the system will display an "**Invalid email or password**" message.

1.6 After successful login, the system will identify the user's role.

1.7 If the user is an **Admin**, the system will redirect them to the **Admin Dashboard**.

1.8 If the user is a **User**, the system will redirect them to the **Resume Analysis Dashboard**.

1.9 When the user clicks the **Logout** button, the system will end the current session and return the user to the login page.

2. Resume Upload and Processing

2.1 Users can upload their resume using a drag-and-drop area or by selecting a file from their computer.

2.2 The system will accept only **PDF (.pdf), Word (.docx), and Text (.txt)** files with a maximum file size of **5 MB**.

2.3 If the uploaded file does not meet these requirements, the system will immediately display an error message.

2.4 After a valid file is uploaded, the system will display a preview of the resume without reloading the page.

2.5 The system will extract the text from the uploaded resume using a document parsing tool.

2.6 When the user clicks **Analyze Resume**, the extracted resume text will be prepared for AI analysis.

3. AI Scoring and Feedback Engine

3.1 Users will enter the target Job Description (JD) together with the uploaded resume.

3.2 The system will send both the resume and job description to an AI model for analysis.

3.3 While the AI is analyzing the resume, the system will display a loading message such as **"Analyzing Resume..."**.

3.4 After the analysis is complete, the system will generate an ATS Match Score between **0% and 100%**.

3.5 The AI feedback will be organized into four categories:

- Skills
- Content Quality
- Structure
- Writing Style

3.6 The system will identify important skills or keywords that are missing from the resume.

3.7 The missing keywords will be displayed as separate recommendation tags to help users improve their resume.

3.8 Users can submit feedback or rate the analysis only after the AI analysis has been completed successfully.

4. User Dashboard and Analytics

4.1 Users can view all previously uploaded resumes, ATS scores, and analysis dates in their dashboard.

4.2 The dashboard will include a **Score Over Time** chart to show the user's progress across multiple resume analyses.

4.3 Users can select a previous analysis to compare the resume and the job description side by side.

5. Admin Panel Management

5.1 After logging in, administrators will access a separate Admin Dashboard.

5.2 The Admin Dashboard will display system statistics such as:

- Total Registered Users
- Total Resumes Analyzed
- Average ATS Score
- Active User Sessions

5.3 Administrators can view a searchable and paginated list of all users with their profile information and roles.

5.4 Administrators can change a user's account status between **Active** and **Banned**.

5.5 Administrators can view system activity logs, AI processing status, and database error logs.

Non-Functional Requirements

NFR-1: Performance

- The system should upload and validate resume files quickly.
 - Resume text should be extracted efficiently.
 - Dashboard information should load without noticeable delay.
 - Resume preview and text extraction should finish within **5 seconds**.
 - AI analysis should normally finish within **15 seconds**.
 - The system should support multiple users at the same time without significant performance issues.
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NFR-2: Security

- Only authorized administrators can access admin features.
- User passwords must be securely hashed before being stored.
- All protected pages must require secure user authentication.

- Unauthorized users must not be able to access admin pages or sensitive data.
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NFR-3: Reliability

- The system should extract resume text accurately and consistently.
 - File processing should not lose or corrupt resume data.
 - Analysis results should remain correct even if unexpected errors occur.
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NFR-4: Usability

- The application should have a clean and consistent Dark theme.
 - Navigation should be simple and easy for both users and administrators.
 - Error messages, warnings, and success notifications should be clear and easy to understand.
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NFR-5: Maintainability

- The system should follow a layered architecture to keep different parts of the application independent.
 - New features such as job recommendations or mock interviews should be easy to add in the future.
 - The source code should be well organized, properly documented, and easy to maintain.
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NFR-6: Scalability

The system should be able to handle growth in:

- Registered users
 - Uploaded resumes
 - Resume analysis history
 - Simultaneous AI analysis requests
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NFR-7: Availability

The system should be available at all times so users can:

- View previous resume analyses
- Upload new resumes
- Receive AI resume analysis whenever needed

NFR-8: Data Integrity

The system should maintain accurate relationships between:

- User accounts
- Uploaded resumes
- AI analysis results

The stored information should remain complete, consistent, and free from corruption.